

Who reads your report, and how?

Starting to write in week 2, not week 10

YOU LEAVE WITH

A plan to 5 February, the four things your marker must believe, and your first 150 words.



Tutorial 2 of 8 • Semester 1

Today's plan

Goal: Know what your report has to do for its reader, and start writing about your project this week.

- 1 The road to 5 February: what you write each week
- 2 Methods: ERA5 cloud, area weighting, anomalies
- 3 How scientists read, and what your marker must believe
- 4 Explain an anomaly to a friend: your first 150 words
- 5 Finding sources and setting up a reference manager

BRING / HAVE READY

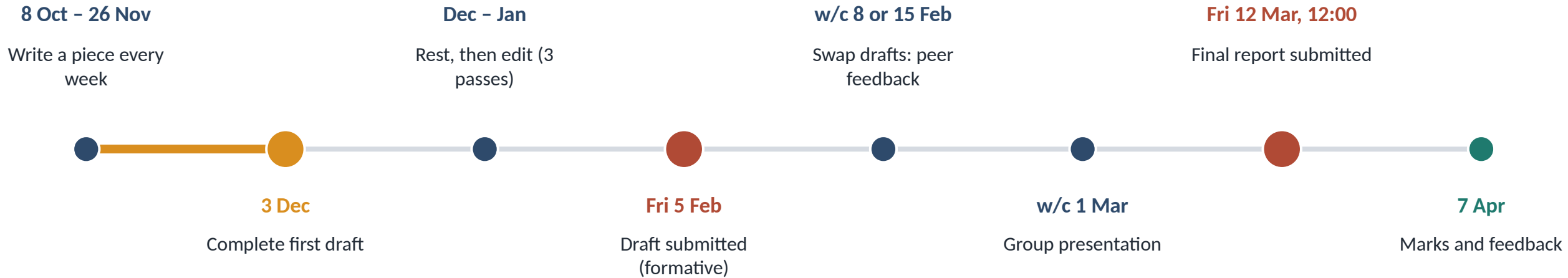
- Your notebook open (Binder or downloaded copy)
- The two patterns you noted in Tutorial 1
- This term's **writing workbook** (Canvas)
- Pen and paper: today you write by hand



The road to 5 February

SEMESTER 1 • THESE TUTORIALS

SEMESTER 2



- Every tutorial from today produces one **piece** of your report; the pieces add up to a full draft on **3 December**.
- January is then for **editing**, not first drafting, and it is also exam season. The draft on 5 Feb carries no marks and has **no DASS extensions**; it exists so you get feedback.
- Report: **max 2,000 words**, individual, 100% of the course mark. AI category: **AI Minimal**.

RIGHT NOW

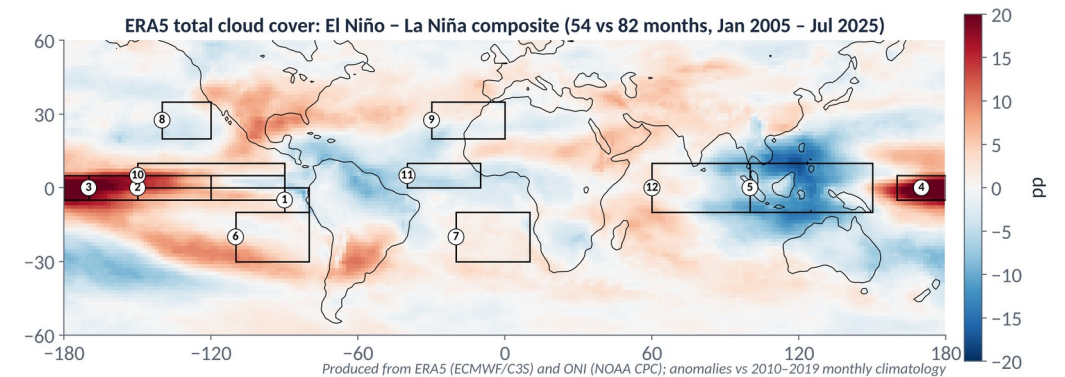
Right now: a very strong El Niño

AS OF LATE SEPTEMBER 2026

- NOAA's Climate Prediction Center has an **El Niño Advisory** in place, and expects a **very strong** event through the northern autumn and winter 2026–27, possibly the strongest since 1950.
- It reports **enhanced convection and rain from the central to the eastern Pacific** and **suppressed convection over Indonesia**: the same pattern our cloud maps show for past El Niños.
- Forecasters at IRI (Columbia) report that all 22 forecast models expect it to strengthen further through the winter.

TWO CAVEATS

- Our dataset ends in **2025**: none of this event is in your analysis.
- Forecasts change: treat this as context, not data.



What past El Niños did to cloud (ERA5, 2005–2025): the shape to watch for this winter.

OPTIONAL: FOLLOW IT LIVE

- Look up this week's **Niño 3.4** value on NOAA CPC's ENSO page.
- Write one sentence: what does it mean for cloud over the central Pacific and the Maritime Continent?

METHODS

What is ERA5 ‘cloud cover’?

- **Reanalysis:** a weather model continually corrected by observations, giving a complete gridded record. Ours: monthly means, January 2005 – August 2025.
- **Cloud is a model product.** ERA5 cloud comes from the model's cloud scheme; observations constrain it only indirectly. Trust the **sign** of a change more than its exact size.
- **Four variables:** total (TCC), low (LCC), medium (MCC) and high (HCC) cloud cover: the **fraction** of each grid box covered (0–1, plotted as %).
- **TCC ≠ LCC + MCC + HCC.** Layers overlap, so seen from above the total is less than the sum.
- Changes are in **percentage points (pp)**: 40% → 44% is +4 pp (a 10% relative increase).

IN YOUR METHODS, THIS BECOMES

“Monthly mean high-cloud cover (HCC) from the ERA5 reanalysis (Hersbach et al. 2020) was analysed for January 2005 – July 2025.”

ONE GRID BOX, SIDE VIEW



Covered when seen from above: TCC = 60%

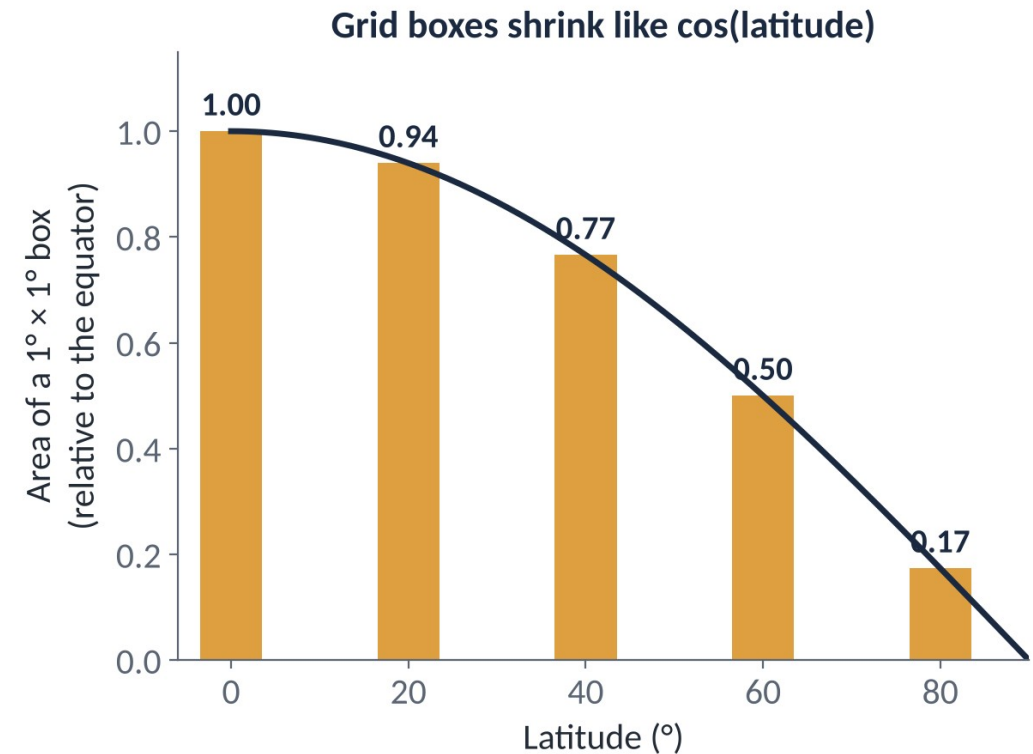
Sum of the layers: $40 + 20 + 30 = 90\%$

Schematic. ERA5 combines layers using an overlap assumption.

METHODS

Averaging over an area: $\cos(\text{latitude})$ weighting

- On a latitude–longitude grid every box spans the same **degrees**, but boxes near the poles cover far less **area**.
- Area $\propto \cos(\text{latitude})$: a box at 60° is **half** the size of one at the equator.
- **Weighted mean:** $\bar{X} = \sum w_i X_i / \sum w_i$, with $w_i = \cos \varphi_i$. An unweighted mean over-counts high latitudes.
- **When it matters:** hugely for a global mean (weights run from 1 to 0); little for a tropical box (5°S – 5°N : 0.996–1.00). Say you did it anyway.
- The notebook weights both global and regional means (area_weights).



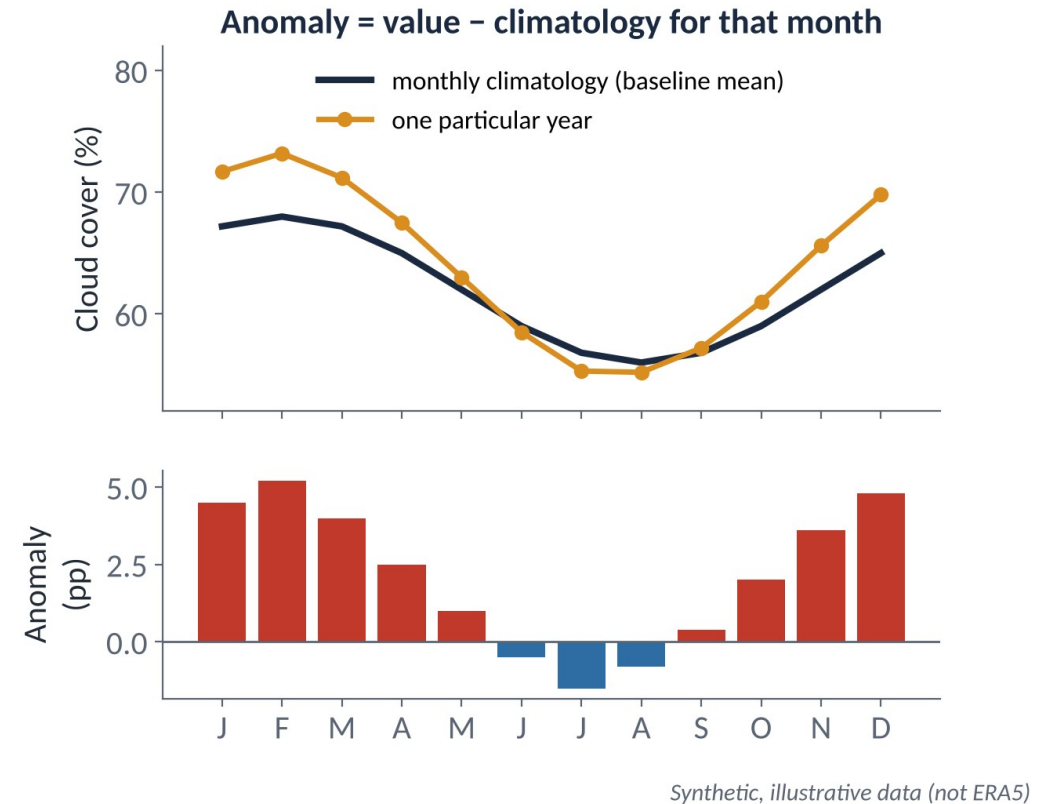
IN YOUR METHODS, THIS BECOMES

“Regional and global means were area-weighted by the cosine of latitude.”

METHODS

Climatology and anomalies

- **Climatology:** the average for each calendar month over a **baseline** period (notebook: 2010–2019, CLIM_START/CLIM_END).
- **Anomaly = value – climatology for that month.** Positive = cloudier than usual **for that place and month**, not ‘very cloudy’.
- Removing the seasonal cycle lets the year-to-year (ENSO) signal show through.
- **The baseline is a choice:** change it and every anomaly shifts. Say which you used.
- Averaging anomalies over a region gives the same answer as taking the anomaly of the regional mean. Both steps are linear.



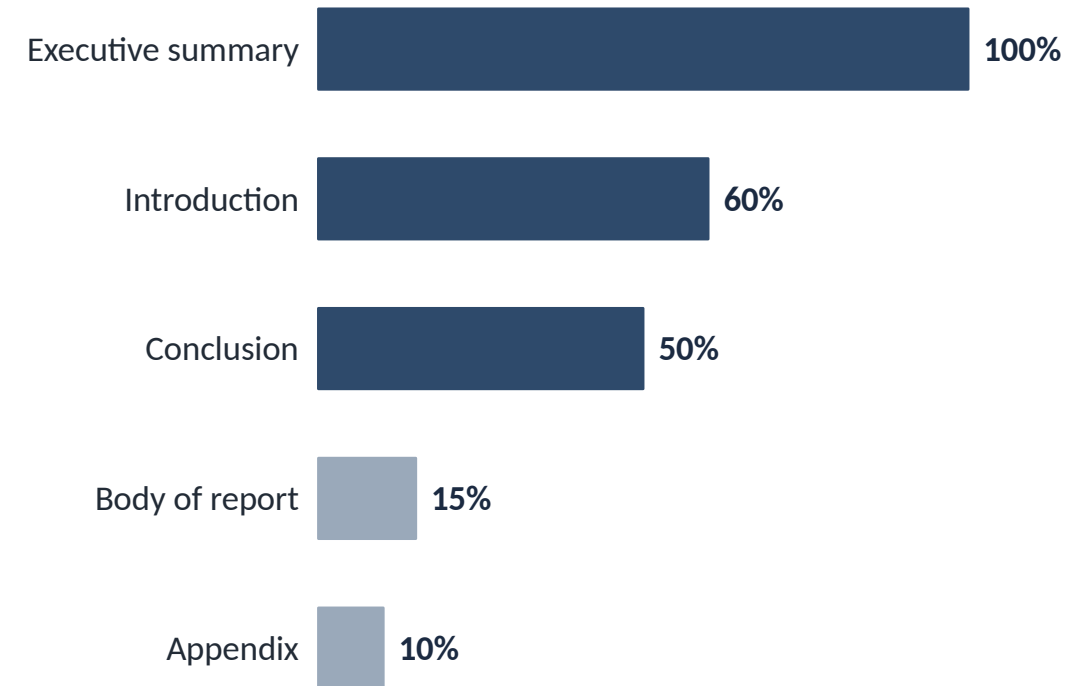
IN YOUR METHODS, THIS BECOMES

“Anomalies were calculated relative to the 2010–2019 monthly climatology.”

How scientists actually read

- **Almost nobody reads a paper front to back.** Title → abstract/introduction → figures → conclusion, and only then (maybe) the rest. Schultz calls this **nonlinear reading**.
- **Your marker reads every word**, but forms a view early from the **title, first paragraph, figures and conclusion**.
- **So put your best, clearest writing:** where the reader looks first, and make each paragraph's **first sentence** carry its point.

Westinghouse managers: share who read each part of a report



Schultz (2009, §4.2): 'Nine reports sat on their shelves largely unread for every one that was read in its entirety.'

Four things your marker must believe

Research funders judge proposals on four propositions (Research Funding Toolkit). A report is judged the same way: only the tense changes.

IMPORTANCE

The question is worth asking.

Introduction: why should anyone care what ENSO does to this cloud, here?

SUCCESS

The report actually answers it.

Results + Conclusion: a direct answer, backed by quantified evidence.

VALUE

Every figure and sentence earns its place.

2,000 words, 3–5 figures: anything that doesn't serve the question goes.

COMPETENCE

The methods are sound and the limits honest.

Methods + Discussion: could someone repeat it, and do you know where it could fail?

AI Minimal: what it means in practice

- **Allowed:** grammar, spelling, fluency and formatting support on text you have already written.
- **Not allowed:** generating or choosing your question, structure, arguments, interpretation or conclusions; summarising papers for you; drafting or rewriting paragraphs.
- **Why it matters here:** the marks are for **your** scientific thinking, and in this course the writing **is** the thinking we are teaching.
- **Protect yourself:** keep **dated versions** (e.g. report_v04_2026-11-12.docx) and your **writing log**. Together they show how your work developed.
- **Unsure whether something is allowed?** Ask your tutor **before** you do it, not after.

THE POLICY

Category: AI Minimal

“AI may be used only for limited presentation-related support, such as grammar, fluency, spelling, or formatting. Students' ideas, arguments, and conclusions must remain their own.”

EART29200 course unit description

IN-CLASS ACTIVITY • 13 MIN

Explain it to a friend

- 1 Imagine writing to a friend who studies something else (Schultz: write 'a letter to a friend about your work in plain language').
- 2 In **150 words, by hand, no notes**, explain (a) what a cloud-cover **anomaly** is and (b) why ENSO might change cloud somewhere.
- 3 Stop at 7 minutes, even if unfinished. Count your words.
- 4 Swap with a neighbour. **Underline** the clearest sentence. **Circle** the first place you had to reread.
- 5 Get yours back. Rewrite the circled sentence in the margin.

YOU LEAVE WITH

- 150 words in your workbook (Task 2A)
- Raw material for your **Methods** (anomaly) and **Introduction** (why ENSO matters for cloud)

PAIR CHECK

- Did they use the word 'anomaly' correctly: unusual **for that month**, not just 'high'?
- Did the ENSO explanation name a **mechanism** (e.g. the Walker circulation), not just 'ENSO affects weather'?

LITERATURE

Finding sources, and keeping track of them

- **Google Scholar:** broad and fast (preprints, theses too) but noisy. Settings → Library links → University of Manchester, so full texts open through your login.
- **Web of Science (formerly 'Web of Knowledge'):** webofscience.com → sign in through the University (institutional login). Curated peer-reviewed journals, precise filters. Start from **one good paper**, go **backward** (references) and **forward** ('Times cited'); filter for reviews.
- **Library Search:** books and full text. Core journals: Journal of Climate, J. Atmos. Sci., Geophys. Res. Lett., Q. J. R. Meteorol. Soc.
- **Background vs evidence:** NOAA Climate.gov is great background; scientific claims need peer-reviewed sources.
- **AI Minimal:** don't use AI tools to find, choose or summarise papers for you.

SET UP A REFERENCE MANAGER THIS WEEK

Zotero (recommended) or EndNote

- **Zotero** (free, zotero.org): the browser connector saves a paper in one click; the Word / Google Docs plugin inserts citations and builds the reference list.
- **EndNote:** licensed by the University (free via the software download service); Harvard Manchester style; Library workshops.
- Avoid starting on **Mendeley Desktop:** legacy, no further updates.
- Pick **one** author-year style and keep it.

Before Tutorial 3 · Thu 15 Oct

About 2½ hours in total. Everything goes in your writing workbook; bring it next week.

2A · 20 min

Explain it to a friend (final)

- Redraft today's piece to **≤200 words**
- Fix the circled sentence
- Add one sentence on **where** you'd look for an ENSO cloud signal

2B · 45 min

Reverse-outline a real paper

- Find one paper on ENSO and clouds (Google Scholar or library)
- One line per Introduction paragraph: its **job** + its point
- Where is the **'however'**? Where is the purpose?

2C · 20 min

Set up: log + reference manager

- **Writing log:** after every notebook session, what I ran / noticed / puzzled me
- Install **Zotero** (or EndNote); save your 2B paper into it
- Choose one author-year citation style

NOTEBOOK STRAND

- Run notebook **sections 1–3** (seasonal cycle, global ENSO maps, global-mean anomalies).
- Analysis slides: **02 Climatology and anomalies** and **03 Finding an ENSO signal**.

READ · SCHULTZ (2009) ELOQUENT SCIENCE

- **§4.1–4.2** (parts of a paper; nonlinear reading)
- **Ch. 7** Accessible scientific writing (≈ 6 pages)

LITERATURE

Searching Web of Science: a quick guide

SEARCH, REFINE, CHASE, SAVE

- **Sign in:** webofscience.com → sign in through the University (institutional login).
- **Advanced Search**, Core Collection. **Fielded Search** tab: one idea per **Topic** row, joined by AND, with **no** TS= tags. **Query Builder** tab: paste a tagged query straight into the **Query Preview** box (TS= topic, TI= title, AU= author, PY= year).
- **Phrases** in quotes ("Walker circulation"); **AND / OR / NOT**; truncation with * (cloud* = cloud, clouds, cloudiness); NEAR/3 for words close together.
- **Refine:** Document Types → Review Article; Web of Science Categories → Meteorology & Atmospheric Sciences; publication years.
- **Sort** by Citations (highest first) to find the classics; by Date for the newest.
- **Chase:** Cited References (backward), Times Cited (forward), Related Records (similar papers).
- **History:** combine searches (#1 AND #2) without retyping.
- **Save:** the Zotero Connector grabs one result or a whole page; or Export → RIS / EndNote.

TRY THIS SEARCH

TS=("El Nino" OR ENSO OR "Southern Oscillation")

AND TS=("cloud cover" OR "cloud fraction" OR "cloud amount")

AND TS=(reanalysis OR ERA5 OR satellite)

Query Builder tab: paste into the **Query Preview** box (not the term box above it). Or use three Topic rows in Fielded Search, minus TS=(). Then refine to Review Article, sort by Citations, and swap in your own region or cloud type.

SCHOLAR OR WEB OF SCIENCE?

- **Scholar** to explore broadly and find full texts fast.
- **Web of Science** for a precise search of peer-reviewed journals, reliable citation counts and filters.

LITERATURE

Zotero in Word: set up once, then three rules

SET UP (15 MIN, ONCE)

- Install **Zotero** (desktop) and the **Zotero Connector** for your browser from [zotero.org](https://www.zotero.org).
- The **Word plugin** installs itself the first time Zotero runs; restart Word and a **Zotero tab** appears (Windows and Mac).
- **Zotero tab: Add/Edit Citation** inserts a citation; **Add/Edit Bibliography** builds the reference list; **Document Preferences** sets the style.
- **Google Docs**: use the Zotero menu that the Connector adds. **LibreOffice** has its own plugin.
- **Word for the web (OneDrive in a browser) is not supported**: open the file in the desktop Word app to cite.
- Optional: a free Zotero account syncs your library between computers (300 MB free storage).

THREE RULES

- **1 Never hand-edit** a citation or the reference list: Zotero overwrites it on the next refresh. Fix the entry in your Zotero library instead.
- **2 One style, chosen once** in Document Preferences (an author-year style, e.g. a Harvard variant or the AMS journal style).
- **3 Share a PDF**, not the live Word file, for the February peer swap, so citations show correctly for someone without Zotero.
- Check every saved item: title, authors, year, journal and DOI. Connectors occasionally grab messy metadata.